ASCII MASTER FLOW — PANEL 1/12: Institutional command map

```
DEPARTMENT OF SPACE -> policy umbrella and sovereign direction
ISRO -> public R&D, launch vehicles, spacecraft and missions
IN-SPACe -> promote, authorise and supervise non-government activity
NSIL -> commercial contracts, services and technology transfer
RULE -> institution names are not interchangeable
```

ASCII MASTER FLOW — PANEL 2/12: Centre-to-launch chain

```
VSSC -> launch-vehicle design and development
LPSC / IPRC -> liquid propulsion and propulsion testing roles
SDSC-SHAR -> integration, range and launch operations
MISSION CONTROL -> flight monitoring and decision chain
RULE -> a centre function is not the whole ISRO mandate
```

ASCII MASTER FLOW — PANEL 3/12: Policy and law stack

```
OUTER-SPACE TREATIES -> international State obligations
INDIAN SPACE POLICY 2023 -> executive role allocation
IN-SPACe NORMS -> authorisation and supervision interface
COMMERCIAL CONTRACT -> service and risk allocation
GAP -> policy is not a comprehensive domestic space statute
```

ASCII MASTER FLOW — PANEL 4/12: Launcher-payload separation

LAUNCH VEHICLE -> supplies ascent energy and guidance
PAYLOAD FAIRING -> protects the mission payload
SEPARATION -> launcher task and spacecraft task diverge
SPACECRAFT -> carries payload and may raise or maintain orbit
RULE -> rocket success and spacecraft service entry are separate

ASCII MASTER FLOW — PANEL 5/12: Orbit vocabulary matrix

```
LEO -> altitude class, often used by crew and Earth observation
POLAR -> high-inclination ground-track geometry
SUN-SYNCHRONOUS -> repeated local-solar-time crossing
GSO -> orbital period tied to Earth's rotation
GEO -> circular equatorial geosynchronous special case
```

ASCII MASTER FLOW — PANEL 6/12: GTO-to-GEO rail

```
LAUNCHER ASCENT -> reaches transfer conditions
GTO INJECTION -> elliptical intermediate orbit
SPACECRAFT PROPULSION -> raises perigee and circularises
GEO SLOT -> circular equatorial final service orbit
TRAP -> GTO delivery is not direct GEO placement
```

ASCII MASTER FLOW — PANEL 7/12: Operational launcher matrix

```
PSLV -> versatile workhorse and mixed mission profiles
GSLV -> cryogenic upper stage and transfer-orbit role
LVM3 -> heavier operational class and human-rating base
SSLV -> separate small-satellite launch architecture
RULE -> role, propulsion, orbit and status must travel together
```

ASCII MASTER FLOW — PANEL 8/12: Lineage and status ladder

```
SLV-3 -> experimental orbital-access learning
ASLV -> intermediate technology-learning step
PSLV / GSLV / LVM3 -> operational families in owner
SSLV -> developed small-launch family with dated status needed
NGLV -> approved future development, not an operational launcher
```

ASCII MASTER FLOW — PANEL 9/12: Propulsion comparison

SOLID -> stored propellant and limited throttle control
EARTH-STORABLE LIQUID -> ambient-temperature liquid pair
CRYOGENIC -> liquid oxygen plus liquid hydrogen
SEMI-CRYOGENIC -> liquid oxygen plus kerosene
RULE -> propellant class does not prove flight status

ASCII MASTER FLOW — PANEL 10/12: Human-rating gate

RELIABLE LAUNCHER -> necessary but insufficient base
MARGINS AND REDUNDANCY -> tolerate selected failures
HEALTH MONITORING -> detect unsafe conditions
ABORT CAPABILITY -> protect crew across ascent phases
CERTIFICATION -> separate human-rated configuration
MUST REMEMBER: India's space architecture separates the Department of Space, ISRO...
MUST REMEMBER: India's space architecture separates the Department of Space, ISRO...

ASCII MASTER FLOW — PANEL 11/12: Access-to-space system

```
VEHICLE -> propulsion, structure, avionics and guidance
FACTORY -> repeatable stage and subsystem quality
PAD / RANGE -> integration, safety and launch window
MISSION ASSURANCE -> verification and anomaly closure
OUTPUT -> cadence and customer confidence
CLOSE DISTINCTION: Organisation is not operation, launcher is not spacecraft, stage is...
```
CLOSE DISTINCTION: Organisation is not operation, launcher is not spacecraft, stage is...

ASCII MASTER FLOW — PANEL 12/12: Future-claim firewall

DEFINE -> institution, vehicle, spacecraft, orbit and service
CLASSIFY -> retired, operational, approved or under construction
TRACE -> propulsion, infrastructure, integration and assurance
SEPARATE -> technology transfer from authorisation and safety
VERIFY -> dated official capacity, schedule and outcome source
EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Date every mission and capability claim;...
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